Maria Montoya-Aguirre

mariamontoyaa.github.io

maria.montoya@psemail.eu
48 Boulevard Jourdan, 75014 Paris

Last update: October 30, 2026

Education

Paris School of Economics & Université Paris 1 Panthéon-Sorbonne
PhD in Economics — 2022 – 2027 (expected)
Supervisor: David Margolis. Thesis committee: Oliver Vanden Eynde, Nina Guyon.
MSc in Analysis and Policy in Economics — 2019 – 2021

University of Chicago, Harris School of Public Policy
Visiting Graduate Student. Host: Chris Blattman. — Fall 2024

Centro de Investigación y Docencia Económicas (CIDE)
BSc in Economics — 2015 – 2019

Fields

Development Economics, Crime, Labor Economics, Political Economy

Research

Job Market Paper

"Drug Lords in STEM"

Education is well established as a protective factor against criminal involvement because it improves legal labor-market opportunities. But what happens when illegal markets demand skilled labor? I study how a boom in chemistry-intensive illegal drug production reshaped human capital investment in Mexico. China's 2019 ban on fentanyl pushed its synthesis into clandestine Mexican laboratories, expanding synthetic drug production. Using a difference-in-differences design, I compare commuting zones with pre-existing synthetic-drug infrastructure to those without before and after the ban. Chemical byproducts of drug synthesis are more often detected in the water of these exposed areas than elsewhere, indicating they are production sites. After the ban, school dropout fell in exposed zones, while time spent studying and the share of students reaching the final year of high school increased. University enrollment also reallocated toward chemistry-related STEM programs, with no detectable change in total enrollment. Illegal production also raised the legal demand for chemistry skills: the wage premium on chemistry-related degrees and employment in chemistry occupations both rose. Meanwhile, among adults whose schooling predates the ban, fewer chemistry degree holders appear in the survey, suggesting that skilled chemists are entering illegal production. As criminal organizations come to rely on skills that legal employers also value, these findings raise the question of how effective education will be at reducing crime.

Publications

"Filling the Gaps: Childcare Laws for Women's Economic Empowerment" with S Anukriti, Lelys Dinarte-Diaz & Alena Sakhonchik. 2025. *AEA Papers and Proceedings* 115: 689–94.

Working papers

"Moral Force, Contagious Force: Partisanship, Leadership, and Public Health Compliance during COVID-19" with ⓡL. Guillermo Woo-Mora & ⓡFederico Daverio-Occhini. November 2025. **Submitted**

"Embracing modernity pays: Cadastre modernization effects on local property tax collection' with ⓡEmmanuel Chávez & ⓡL. Guillermo Woo-Mora. November 2024.

"Drug-led development? Organized crime and local employment in Mexico"

Selected work in progress

"Evaluating Demand- and Supply-Side Interventions Against Organized Crime Recruitment" with Natalia Guerrero-Trinidad & Maria Micaela Sviatschi.

"Exporters into Retailers: Foreign Interdiction and the Formation of Drug Markets around Schools" with Natalia Guerrero-Trinidad

"Deter or displace? Securing schools against organized crime in Ecuador" with Natalia Guerrero-Trinidad

"From Labs to Rivers: Environmental and Health Consequences of Mexico's Fentanyl Boom" with Sandra Aguilar-Gómez.

"DICE-MX: A Georeferenced Event-Level Dataset of Drug Interdiction and Criminal Events in Mexico"

Relevant Experience

Research and policy

Research Assistant – Princeton University RPDE 2025–2026

Research Assistant – The World Bank (Development Research Group) 2021–2024.

Research Analyst – United Nations Development Programme 2021–2022.

Junior Analyst – Dateras México 2018–2019

Teaching

Teaching Assistant – Sciences Po 2025. Public Economics (Masters in Public Policy).

Instructor – Paris School of Economics 2023, 2025. Introduction to Statistical Programming in R (Masters in Analysis and Policy in Economics).

Grants & Awards

Award: Young Economist Award 2026 – European Economic Association

Award: Selected for the EALE Tour 2027

Award: Joaquín Xirau Icaza Economics Prize 2026 – El Colegio de México

Research grant: Citizen Security Research Initiative 2026 – Innovations for Poverty Action (PI, USD 72,532).

Research grant: Citizen Security Research Initiative 2026 – Innovations for Poverty Action (co-PI, USD 70,861).

Research grant: Dissertation Grant 2026 – Horowitz Foundation for Social Policy (USD 10,000).

Award: Honorable Mention, Premio Banamex de Economía 2024 (with L. Guillermo Woo-Mora & Federico Daverio-Occhini).

Award: Honorable Mention, Victor L. Urquidi Prize in Economics 2024 - El Colegio de México (with Emmanuel Chávez & L. Guillermo Woo-Mora).

Research grant: Decentralización y Capacidades Subnacionales en América Latina y El Caribe 2023 – CAF (USD 15,000).

Research grant: Fieldwork Grant 2023, Paris School of Economics (EUR 1500)

Scholarship: Paris 1 Panthéon-Sorbonne ED465 PhD scholarship 2022-2025

Award: National Public Finance Prize 2020, First Place – CEFP, Mexican Chamber of Deputies (with L. Guillermo Woo-Mora and Eduardo Ortiz-Juárez)

Scholarship: Eiffel Excellence Scholarship 2020, Government of France.

Scholarship: APE Scholarship 2019, Paris School of Economics.

Conferences and Seminar Presentations

2026: *CEPR Development Economics Annual Symposium 2026-1*, Trinity College Dublin; *7th Annual Texas Economics of Crime Workshop*, University of Texas-Austin; *Princeton Workshop on Organized Crime Research and Policy*, Princeton SPIA in DC; *LSE-Chicago Conference on the economics of crime and justice*, London School of Economics; *5th Workshop on the Economics of Crime for Junior Scholars*, Institute for Fiscal Studies; ViCE Seminar; *EEA-ESEM 2026*, University College Dublin; *EALE Conference*, Universitat de Barcelona & BSE; Princeton Development Economics Student Workshop; NEUDC*; *LACEA-LAMES**

2025: *ASSA Annual Meeting*; *BSE Summer Forum (Public Economics)*.

2024: *LACEA–LAMES*, Universidad de la República Montevideo; *Mini-conference on the Economics of Crime*, University of Chicago

2023: *XVII RIDGE Forum & AL CAPONE Workshop on the Economics of Crime*, CAF Montevideo

* Scheduled presentations.

Refereeing

Journal of Development Economics, World Development.

Skills

Software: R, Python, Stata, QGIS.
Languages: Spanish (native); English (fluent); French (advanced).

References

David Margolis
Paris School of Economics & CNRS
Professor · Senior Researcher
david.margolis@psemail.eu

Chris Blattman
University of Chicago, Harris School of Public Policy
Professor
blattman@uchicago.edu

Maria Micaela Sviatschi
Princeton University
Associate Professor
msviatschi@princeton.edu

Nina Guyon
Paris School of Economics
Associate Professor
nina.guyon@psemail.eu

Sylvie Lambert
Paris School of Economics & INRAE
Professor · Senior Researcher
sylvie.lambert@psemail.eu